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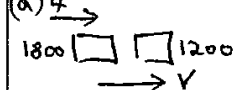
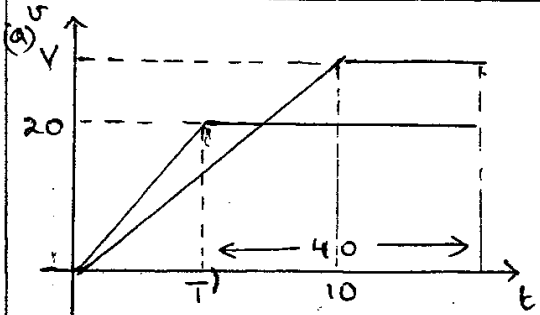
January 2002

Advanced Supplementary/Advanced Level

General Certificate of Education

Subject MECHANICS 6677

Paper No. M1

Question number	Scheme	Marks
1.	$\text{Impulse} = \text{change in mom}^m = 0.3(8+6)$ $= \underline{4.2 \text{ N s}}$	M1 A1 A1 (3)
2.	(a)  $1800 \cdot 4 = (1800 + 1200) V$ $V = \underline{2.4 \text{ m s}^{-1}}$ (b) $R \cdot 8 = 3000 \cdot 2.4$ $R = \underline{900}$	M1 A1 A1 (3) M1 A1 $\sqrt{V}$ A1 (3) (6)
3.	(a) "$v = u + at$": $60 = 12 + 4a \rightarrow a = \underline{12 \text{ m s}^{-2}}$ * (b) "$s = ut + \frac{1}{2}at^2$" $0A = 12 \cdot 4 + \frac{1}{2} \cdot 12 \cdot 4^2$ $= \underline{144 \text{ m}}$ (c) "$v^2 = u^2 + 2as$" $v^2 = 12^2 + 2 \cdot 12 \cdot 72$ $v = \underline{43.3 \text{ m s}^{-1}}$	M1 A1 (2) M1 A1 A1 (3) M1 A1 $\sqrt{0A}$ A1 (3)
4.	(a)  One shape correct 2nd shape correct rel. to first Figs (10, 20, 40) <u>1 contd.</u>	B1 B1 B1 (3)

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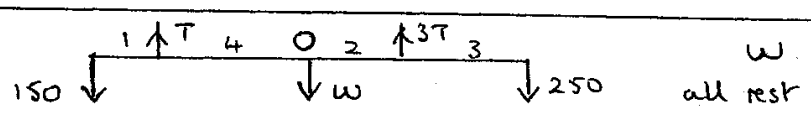
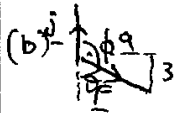
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4.	(b) Scooter: dist travelled = area under graph $850 = \frac{1}{2} T \cdot 20 + 20 \cdot 40$ $\Rightarrow T = \underline{5s}$ (c) Van: $850 = \frac{1}{2} V \cdot 10 + V(40 - 5)$ $\Rightarrow V = \underline{21.25 \text{ m s}^{-1}}$	M1 A1 A1 (3) M1 A1 $\sqrt{(T)}$ A1 (3) (9)
5.	(a)  (b) M(O): $150 \cdot 5 + 3T \cdot 2 = T \cdot 4 + 250 \cdot 5$ Solve $T = \underline{250 \text{ N}}$ [Allow M1 A2, 1, 0 for moments eqn² abt any pt. Then M1 A1 for complete sol² $\rightarrow T =$]. (c) R($\uparrow$) $4T = 400 + W \rightarrow W = \underline{600 \text{ N}}$ (M1 needs complete sol² $\rightarrow W =$). (d) By having weight act at <u>centre/mid-pt.</u>	B1 B1 (2) M1 A2, 1, 0 $\downarrow$ M1 A1 (5) M1 A1 (2) B1 (1) (10)
6.	(a) $\underline{F} = (6\hat{i} + 2\hat{j}) + (3\hat{i} - 5\hat{j}) = \underline{(9\hat{i} - 3\hat{j}) \text{ N}}$ (b)  $\tan \theta = \frac{9}{3} \Rightarrow \theta = 71.6^\circ$ $\phi = \underline{108.4^\circ}$ (c) "$\underline{F} = m\underline{a}$" $\Rightarrow \underline{a} = \underline{(3\hat{i} - \hat{j}) \text{ m s}^{-2}}$ (d) $\underline{v} = (-2\hat{i} + \hat{j}) + 2(3\hat{i} - \hat{j}) = \underline{4\hat{i} - \hat{j}}$ $\text{Speed} = \sqrt{4^2 + 1^2} = \underline{4.12 \text{ m s}^{-1}}$	B1 (1) M1 A1 $\sqrt{(F)}$ A1 (3) M1 A1 $\sqrt{(F)}$ (2) M1, M1, A1 $\sqrt{(a)}$ M1 A1 (5) (11)

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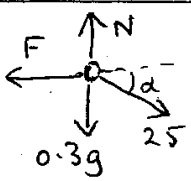
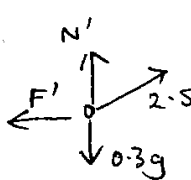
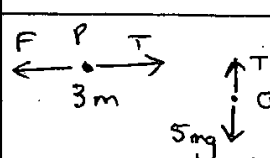
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7.	(a)  $R(\uparrow) \quad N = 0.3 \times 9.8 + 2.5 \sin \alpha$ $(= 2.94 + 1.5 = 4.44 \text{ N})$ $R(\rightarrow) \quad F = 2.5 \cos \alpha \quad (= 2 \text{ N})$ $F = \mu N \rightarrow \mu = \frac{2}{4.44} \approx \underline{0.45}$ (b)  $N' = 0.3 \times 9.8 - 2.5 \sin \alpha = \underline{1.44 \text{ N}}$ $F' \leq \mu N' \quad N' < N \Rightarrow F'_{\text{max}} \text{ less}$ But $F'_{\text{max}} = 2.5 \cos \alpha$ for equilib. Hence equilib. <u>not</u> possible	M1 A2, 1, 0 M1 A1 M1 M1 A1 (8) M1 A1 (2) M1 A1 C50 (2) (12)
8.	(a)  $P: T - F = 3ma$ $Q: 5mg - T = 5ma$ (b) $F = 0.6 \times 3mg \quad (= 1.8mg)$ Hence $5mg - 1.8mg = 8ma$ $a = \underline{0.4g}$ (c) Sub: $T = 3ma + F$ or $5mg - 5ma$ $\rightarrow T = \underline{3mg}$ (d) Speed when Q hits floor: $v^2 = 2 \times 0.4g \times h$ $= \frac{4}{5}gh$ Decel² of P: $3mf = 1.8mg \Rightarrow f = 0.6g$ Dist moved by P: $\frac{4}{5}gh = 2 \cdot \frac{3}{5}g \cdot s$ $\Rightarrow s = \underline{\frac{2}{3}h}$	M1 A1 M1 A1 (4) M1 A1 M1 A1 (4) M1 A1 (2) M1 A1 ✓ M1 A1 M1 A1 (6) (16)